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DC Motor Thermal Diagnostic & Maintenance Checklist

Shop-floor SOP for maintenance engineers and technicians: isolate the heat source, verify it, fix it

Work through Step 1 before anything else: whether the motor overheats with the load connected or also when uncoupled decides which of the later steps apply. Record every reading so you have a baseline for the next inspection.

Step 1 — Establish the facts (motor running, safe distance)

- [ ] Read the nameplate: rated voltage, full-load current (FLA), rated speed, duty type (S1/S2/S3) and insulation class.
- [ ] Measure frame temperature with an IR thermometer at mid-stator and at both end shields; note the ambient temperature 150 mm from the air inlet.
- [ ] Measure armature current under the heaviest normal load with a DC clamp meter; compare with FLA (investigate above 100 %, act above 105 % continuous).
- [ ] Measure terminal voltage under load with a true-RMS meter; deviation beyond ±5 % of nameplate needs correcting.
- [ ] Note how long the motor runs before it gets hot and how many starts per hour the machine makes.

Step 2 — Loaded vs no-load test (isolate and lock out before uncoupling)

- [ ] Uncouple the motor from the driven machine and run it at rated voltage for 20–30 minutes.
- [ ] Runs cool uncoupled → the heat comes from the load side: binding, overload, wrong gear ratio, belt over-tension. Go to Step 3.
- [ ] Still overheats uncoupled → the fault is in the motor or its supply: armature short, overvoltage, ripple, bearings or brushes. Go to Steps 4–6.
- [ ] Turn the shaft by hand (power isolated): it should rotate freely with no rough spots, binding or axial play.

Step 3 — Load-side and duty checks

- [ ] Inspect the driven machine for misaligned sprockets, seized rollers, dry gearboxes and over-tightened belts.
- [ ] Check shaft alignment: total indicated runout within 0.05 mm.
- [ ] Compare the duty type on the nameplate with the logged run/rest pattern; an S2 or S3 motor must not run continuously.
- [ ] If continuous load exceeds about 80 % of rated torque, add gear reduction or move to a larger frame.
- [ ] For frequent starts, add a soft-start ramp to limit inrush current.

Step 4 — Supply quality checks

- [ ] Check that the supply voltage under load is within ±5 % of nameplate; correct the supply tap or cable size if not.
- [ ] Check the waveform with an oscilloscope; a form factor (I rms / I avg) well above 1.05 indicates ripple heating — add a DC choke or filter, or use a PWM drive.
- [ ] Check for voltage drop across long cable runs at full current.
- [ ] Confirm transient suppression (varistor / snubber) is fitted where contactors switch the motor.

Step 5 — Brushes, commutator and bearings (power isolated)

- [ ] Commutator surface: even chocolate-brown film; no burnt bars, grooving, threading or copper drag.
- [ ] Mica undercut 0.8–1.2 mm below the copper; clean carbon dust out of the slots.
- [ ] Brush spring pressure typically 15–20 kPa; replace brushes at one-third of their original length.
- [ ] Bearing end shields hotter than the stator frame → lubrication failure or misalignment; replace with sealed C3-clearance bearings.
- [ ] Clean intake screens and frame fins; even a 1 mm dust layer insulates the motor.

Step 6 — Winding insulation tests

- [ ] Insulation resistance to frame with a megohmmeter: below 1 MΩ means serious insulation damage — do not return the motor to service.
- [ ] Bar-to-bar resistance around the commutator with a micro-ohmmeter: one low reading pinpoints a shorted coil.
- [ ] A burnt-varnish smell or rapid heating at no load points to an inter-turn short — rewind (with VPI) or replace the armature.
- [ ] Record all readings against the motor serial number for trend analysis.

Thermal benchmark by insulation class (40 °C ambient)

Insulation class	Max hot-spot	Max rise (ΔT)	Typical safe frame temp	Investigate frame temp above
Class B	130 °C	80 K	65–75 °C	80 °C
Class F	155 °C	105 K	80–90 °C	95 °C
Class H	180 °C	125 K	95–110 °C	115 °C

Frame temperature typically runs 15–25 °C below the internal winding hot-spot. Every 10 °C above the insulation rating roughly halves insulation life. Values are typical; always confirm against the motor's nameplate and test sheet.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.